

Mechanism Design with Multiple Principals

Outline

Competing Hierarchies (Myerson, 1982)

- $\mathcal{K}$ is the set of principals, $\mathcal{K} = \{1, 2, \dots, K\}$, with $k \in \mathcal{K}$;
- Each principal k interacts with n_k agents only, and we denote the typical agent of principal k as $i^k \in \mathcal{N}^k$. We assume that each agent belongs only to one corporation (the sets $\mathcal{N}^k$ are disjoint).

Following Myerson (1982), we call the collection of a principal and his agents, $\{k\} \cup \mathcal{N}^k$, *corporation* k .

BIP with Multiple Principals

The Bayesian incentive problem is the structure $(\mathcal{K} \cup \mathcal{N}, D, A, \Theta, p(t), U)$, where

- $D^k \times A^k = D^k \times A_1^k \times \dots \times A_{N_k}^k$ is the (finite) set of all possible decisions in corporation k
- An allocation is a vector in $D \times A = \times_k (D^k \times A^k)$;
- We call $\Theta^k = \Theta_1^k \times \dots \times \Theta_{N_k}^k$ the (finite) set of types for agents of principal k and $\Theta = \times_k \Theta^k$;

BIP with Multiple Principals (cont'd)

- $p(\theta)$ is the common prior, i.e. a commonly known probability distribution over the types that each player uses to compute his/her beliefs about other players' types;
- $U^k = (U_1^k, \dots, U_{n_k}^k)$ where $U_i^k : D \times A \times \Theta \rightarrow \mathbb{R}$ for $i = 0, 1, \dots, n_k$ is the vector of the agents' payoff functions in corporation k .
- Similarly, we denote $V^k : D \times A \times \Theta \rightarrow \mathbb{R}$ the payoff to principal $k \in \mathcal{K}$

Coordination Mechanism with Multiple Principals

Each principal $k \in \mathcal{K}$ may design a *coordination mechanism* $((R^k, M^k), q^k)$, where

- $M^k = \prod_{i \in \mathcal{N}^k} M_i^k$ is the set of messages that can be sent by the agents in corporation k and $R^k = \prod_{i \in \mathcal{N}^k} R_i^k$ is the set of recommendations that the principal can send to his agents;
- $q^k : D^k \times M^k \times R^k \rightarrow \mathbb{R}_+$ is a probability function, assigning a probability to each combination of messages and principal's decision, conditional on the reports received from the agents.

A (pure) strategy for agent i^k strategy is the pair $\rho_i^k : \Theta_i^k \rightarrow M_i^k$ and $\sigma_i^k : R_i^k \times \Theta_i^k \times \rightarrow A_i^k$.

Direct Mechanism

A *direct* mechanism for principal k is a conditional probability function of the form

$$q^k : D^k \times A^k \times \Theta^k \rightarrow \mathbb{R}_+$$

$\Rightarrow q^k(d^k, a_i^k | t^k)$ is the probability that principal k takes the decision d^k and recommends the action a_i^k to his agent i , for all $i \in \mathcal{N}^k$, given that she received report of types $\theta^k \in \Theta^k$.

Observe that the payoffs of a principal-agents corporation are affected by the actions of players in other corporations.

$\Rightarrow$ We cannot define incentive-compatibility for only one corporation in isolation, but we must take into account *all* decisions in society.

Incentive-Compatible Mechanisms

Definition 4

A direct mechanism q^k is Bayesian incentive-compatible given the mechanisms $(q^1, \dots, q^K)$ if the honest-obedient strategies $(\rho_i^k, \sigma_i^k)_{i \in \mathcal{N}^k}$ form an equilibrium when principal k posts the mechanism q^k , and:

- each other principal j ($j \neq k$) posts the mechanism q^j*
- the agents for every other principal j are expected to be honest and obedient*

Incentive-compatible Mechanisms (cont'd)

In particular, any agent i of corporation k must not have an incentive to unilaterally deviate from the honest and obedient behavior once she learns her type:

$$\mathbb{E}[U_i^k(d, a, t) | \theta_i^k] = \sum_{\theta_{-i} \in \Theta_{-i}} \sum_{d \in D} \sum_{a \in A} p(\theta_{-i} | \theta_i^k) q(d, a | \theta) U_i^k(d, a, \theta) \geq$$

$$\sum_{\theta_{-i}} \sum_{d \in D} \sum_{a \in A} p(\theta_{-i} | \theta_i^k) q^{-k}(d^{-k}, a^{-k} | \theta^{-k}) q^k(d^k, a^k | \theta_{-i}^k, \hat{\tau}_i^k) U_i^k(d, (a^{-k}, a_{-i}^k, \hat{\delta}_i^k(a_i^k)), \theta) \quad (IC)$$

for all $i \in \mathcal{N}^k$, $\theta_i^k \in \Theta_i^k$, $\hat{\tau}_i^k \in \Theta_i$ and $\hat{\sigma}_i^k : A_i^k \rightarrow A_i^k$, where

$$q(d, a | \theta) = \prod_{j \in \mathcal{K}} q^j(d^j, a^j | \theta^j) \quad \text{and} \quad q^{-k}(d^{-k}, a^{-k} | \theta^{-k}) = \prod_{j \in \mathcal{K} \setminus \{k\}} q^j(d^j, a^j | \theta^j)$$

Incentive-compatible mechanisms (cont'd)

We focus on direct mechanisms.

⇒ The problem of identifying the optimal mechanism can still be formulated as a constrained optimization problem for principal k , taking as given the other principals and agents' behaviors, i.e. for any given $(q^j)_{j \in \mathcal{K} \setminus \{k\}}$.

However, now principal k 's optimal incentive compatible mechanism depends on the mechanism chosen by the other principals.

Principals' Equilibria

Definition 5 (Principals' equilibrium)

*The vector $(q^k)_{k \in \mathcal{K}}$ is a **principals' equilibrium** if, for each principal $k \in \mathcal{K}$, q^k maximizes the principal's expected utility subject to (IC), given the other q^j , for all $j \neq k$, where q^k is a valid probability function.*

Applications: Competing Hierarchies

Manufacturers who produce intermediate goods to be sold to retailers for trade in the consumption markets

- 1 Retailers hold private information on market conditions and each of them is exogenously assigned to only one manufacturer.
- 2 Strategic interactions between retailers arise because the demand price faced by a retailer of hierarchy j depends on the final output of all hierarchies.
- 3 Each manufacturer j proposes a direct mechanism that assigns a price-quantity pair $(p_j(\theta), q_j(\theta))$ to each type of retailer of his hierarchy who truthfully reports its type θ .

Applications: Competitive Search

Privately informed workers apply to firms who face capacity constraints, and may therefore end up being rationed.

- ① Firms compete by offering price-quantity contracts and workers apply to at most one firm. A single worker can only communicate with a firm when applying for its vacancies.
- ② For a given application decision and a given type, a worker's payoff depends on the contract posted by the firm she applies for and on the rationing probability
- ③ The ratio between the number of principals posting a given mechanism and the number of agents applying for it determines the rationing probability which every agent takes as given.

Non-Existence of a Principals' Equilibrium

Proposition 1 (*Non-Existence of a Principals' Equilibrium*)

Myerson (1982): Principals' equilibria do not always exist.

Proof of Proposition 1

To establish the result, Myerson (1982) provides an example of non-existence of a principals' equilibrium.

Consider this case:

- Two principals ($K = 2$);
- Each principal has one agent ($n_k = 1$, $k = 1, 2$);
- Principals' decisions: $D^k = \{A, B, C\}$ for $k = 1, 2$;
- Agents' actions are observable, i.e. $|A_1^1| = |A_1^2| = 1$;
- $\Theta_1^1 = \Theta_1^2 = \{\alpha, \beta\}$ and types are identically and independently distributed according to a discrete uniform distribution, i.e.
 $P(\theta^k = \alpha) = P(\theta^k = \beta) = \frac{1}{2}$ for $k = 1, 2$.

Proof of Proposition 1 (cont'd)

Payoffs are given in the following table, where each pair contains the payoff of principal k (V^k) and her agent (U_1^k).

V^k, U^k	$\theta^k = \alpha$	$\theta^k = \beta$
$d^k = A$	6, 1	0, z^k
$d^k = B$	0, z^k	6, 1
$d^k = C$	5, 0	5, 0

The term z^k is determined by the other principal's decisions as follows:

- For $k = 1 \rightarrow z^1 = 2$ if principal 2 chooses decisions A or B and $z^1 = 1$, if principal 2 chooses C ;
- For $k = 2 \rightarrow z^2 = 1$ if principal 1 chooses decisions A or B and $z^2 = 2$, if principal 1 chooses C .

Proof of Proposition 1 (cont'd)

W.l.o.g., let us first consider the game from the perspective of $P1$. If $P2$ commits to playing $\{C\}$, then the game reduces to:

V^1, U^1	$\theta^1 = \alpha$	$\theta^1 = \beta$
$d^1 = A$	6, 1	0, 1
$d^1 = B$	0, 1	6, 1
$d^1 = C$	5, 0	5, 0

$\Rightarrow$ There is an incentive-compatible mechanism yielding $P1$ a payoff of 6

This mechanism implements A if the reported type is α and B if β .

Proof of Proposition 1 (cont'd)

Consider now the the game from the perspective of principal 2. If principal 1 sets the above mechanism, then the game reduces to:

V^2, U^2	$\theta^2 = \alpha$	$\theta^2 = \beta$
$d^2 = A$	6, 1	0, 1
$d^2 = B$	0, 1	6, 1
$d^2 = C$	5, 0	5, 0

$\Rightarrow$ Again, the IC mechanism implementing A if the reported type is α and B if β yields $P2$ a payoff of 6.

In this situation is optimal for $P1$ to instead induce $\{C\}$

Proof of Proposition 1 (cont'd)

This observation allows Myerson (1982) to claim that:

- There is no principals' equilibrium (Proposition 3)

A principals' equilibrium is a PBE (γ, λ) , in which, for every j :

$$V_j(\gamma, \rho) \geq V_j(\gamma'_j, \gamma_{-j}, \rho') \quad \forall \gamma'_j \in \Gamma_j, \forall \rho',$$

Interpretation: each principal believes that, in every subgame, agents will coordinate on his preferred continuation equilibrium.

An Alternative View

For each subgame induced by the mechanisms $\gamma = (\gamma_1, \gamma_2)$,

- ① Each type α of each agent k , whenever indifferent between different reports to principal k , picks an $m_k^\alpha \in \{\alpha, \beta\}$ which maximizes $\gamma_k(B|m^k)$
- ② Each type β of each agent k , whenever indifferent between different reports to principal k , picks an $m_k^\beta \in \{\alpha, \beta\}$ which maximizes $\gamma_k(A|m^k)$.

We now show that no principal has a profitable deviation from $\gamma^* = \{C\}$, starting with principal 1 (P1). Let γ_1 be a deviating mechanism for P1, and consider the behavior of agent 1 (A1) in the subgame (γ_1, γ^*) .

An Existence Result

It is immediate to see that each type of A1 is indifferent between the reports α and β to be sent to P1 if and only if:

$$\gamma_1(A|\alpha) + \gamma_1(B|\alpha) = \gamma_1(A|\beta) + \gamma_1(B|\beta). \quad (1)$$

Suppose that γ_1 satisfies (??). Then, we consider two situations in turn. If

$$\gamma_1(A|\alpha) \leq \gamma_1(A|\beta) \iff \gamma_1(B|\alpha) \geq \gamma_1(B|\beta), \quad (2)$$

then, given (i) – (ii), both types of A1 are truthful to P1.

An Existence Result

The corresponding payoff to P1 is

$$\frac{1}{2}(6\gamma_1(A|\alpha) + 5\gamma_1(C|\alpha)) + \frac{1}{2}(6\gamma_1(B|\beta) + 5\gamma_1(C|\beta)) = \quad (3)$$

$$= 5 + \frac{1}{2}(\gamma_1(A|\alpha) - 5\gamma_1(A|\beta)) + \frac{1}{2}(\gamma_1(B|\beta) - 5\gamma_1(B|\alpha)) \leq 5, \quad (4)$$

since $\gamma_1(A|m) + \gamma_1(B|m) + \gamma_1(C|m) = 1$, for each $m \in \{\alpha, \beta\}$. The deviation is therefore not profitable. If, instead, (??) is satisfied while (??) is not, we get

$$\gamma_1(A|\alpha) > \gamma_1(A|\beta) \iff \gamma_1(B|\alpha) < \gamma_1(B|\beta), \quad (5)$$

in which case no type of A1 is truthful to P1.

An Existence Result

The corresponding payoff to P1 is then

$$\frac{1}{2}(6\gamma_1(A|\beta) + 5\gamma_1(C|\beta)) + \frac{1}{2}(6\gamma_1(B|\alpha) + 5\gamma_1(C|\alpha)) = \quad (6)$$

$$= 5 + \frac{1}{2}(\gamma_1(A|\beta) - 5\gamma_1(A|\alpha)) + \frac{1}{2}(\gamma_1(B|\beta) - 5\gamma_1(B|\alpha)) < 5, \quad (7)$$

which, again, makes the deviation not profitable.

Finally, we have to consider the case in which (??) does not hold. Then, neither type of A1 is indifferent, and both types will send the same message, say m , to P1. P1's payoff is then:

$$\frac{1}{2}(6\gamma_1(A|m)+5\gamma_1(C|m))+\frac{1}{2}(6\gamma_1(B|m)+5\gamma_1(C|m)) = \frac{1}{2}(6+4\gamma_1(C|m)),$$

which is smaller than 5 for each $\gamma_1(C|m) \in [0, 1]$.

The absence of profitable deviations for P2 can be shown along similar lines.

Equilibrium Existence

In the Myerson (1982) example, existence of a PBE can be restored.

Do we have general existence results?

Only in the single-agent case (Carmona-Fajardo 2009)

$I \geq 2$: we are still working on it !

Common Agency and the Delegation Principle (Martimort-Stole, 2001)

Principals interact in the presence of a single agent.

Every principal independently communicates with the single agent selecting a communication process or mechanism. A mechanism chosen by principal i consists of a measurable mapping

$$\gamma_j : M_j \rightarrow \Delta(D_j)$$

We introduce the richness assumption $\#\Theta \leq \#M_i$ for any i .

Communication

Final outcomes are determined in the following way.

- 1— Every principal simultaneously selects and publicly commit to a mechanism
- 2— In a next step, the agent sends a message $m_k \in M_k$ to each principal k and takes her actions.

Participation

CAN BE INCLUDED EASILY

Message Spaces

For a given array of message spaces $M = (M_1, M_2, \dots, M_N)$, the common agency game is:

$$G^M = \{\Theta, \Gamma, \Sigma, V, (U_i)_{i \in \mathcal{N}}\},$$

and we focus on the set of Perfect Bayesian Equilibria (*PBE*) of G^M .

The message spaces can be quite large; one simple game is one where $M_k = \Theta_k$, for every k .

The “Direct Mechanism” Game

Given Θ , the game is given by the tuple:

$$G^{\Theta} = \left\{ \Theta, \tilde{\Gamma}, \tilde{\Sigma}, V, (U_i)_{i \in \mathcal{N}} \right\}.$$

Direct mechanisms have an obvious appeal - the message spaces are simple.

The question then is, when can one restrict attention to direct mechanisms?

Common Agency and the Revelation Principle

Revelation Principle: For any equilibrium $(\gamma^*, \rho^*, \sigma^*)$ of any game G^M there is an equilibrium $(\tilde{\gamma}, \tilde{\rho}, \tilde{\sigma})$ of the game G^Θ that gives rise to the same distribution over D and A and has truth-telling as an equilibrium behavior (Martimort-Stole 2002, p. 1662).

For competing mechanism (and common agency) games the result cannot in general be established.

A “Failure” of the Revelation Principle

Let $K = 2$, $N = 1$, $\Theta = \{\theta\}$, and $A = \{a\}$

Take also $D^1 = D^2 = \{A, B, C\}$. Payoffs are:

	A	B	C
A	$(1, 1, 1)$	$(2, 0, 2)$	$(-1, 5, 10)$
B	$(1, 2, 2)$	$(1, 1, 1)$	$(0, 0, 0)$
C	$(5, -1, 10)$	$(0, 0, 0)$	$(0, 0, 0)$

A “Failure” of the Revelation Principle (cont’d)

Consider first the game G^Θ : a direct mechanism is a take-it-or-leave-it offer for each principal.

The game admits only one (pure strategy) equilibrium outcome, i.e. both principal posting $\{C\}$.

The corresponding allocation is $(0, 0, 0)$ which is inefficient.

A “Failure” of the Revelation Principle (cont’d)

Take $M_1 = M_2 = \{m_b, m_c\}$, and consider the mechanisms

$$\gamma_k = \begin{cases} d_k = B & \text{if } m_k = m_b \\ d_k = C & \text{if } m_k = m_c \end{cases}$$

The strategies $\{\gamma_1, \gamma_2\}$ together with the messages $\{m_b, m_b\}$ form an equilibrium of the game Γ_M . The corresponding outcome is (B, B) .

The Delegation Principle

A new canonical game. Let $\hat{G}$ be the “Menu Game”:

- 1- Principals compete by simultaneously posting menus. A menu $\hat{\gamma}_k$ is any subset of the set of feasible decisions D_k .
- 2- The agent chooses an item from each principal's menu and takes a (non contractible) action $a \in A$.

In this game, the strategy space of each principal k is $\hat{\Gamma}_k = 2^{D_k}$

The Delegation Principle: Back to the Example

- Consider $\hat{G}$: the allocation $(1, 1, 1)$ can be supported at equilibrium if $\hat{\gamma}_1 = \hat{\gamma}_2 = \{B, C\}$.
- It cannot, though, be supported at equilibrium in the simpler game in which only take-it or leave-it offers are allowed.
- $(0, 0, 0)$ is an equilibrium allocation of $\hat{G}$.

Delegation Principle: an Informal Statement

Martimort-Stole (2002), Peters (2001): For each (γ^*, σ^*) of every communication game G_M , there exists an equilibrium $(\hat{\gamma}, \hat{\sigma})$ of the corresponding $\hat{G}$ game that gives rise to the same distribution over $D \times A$.

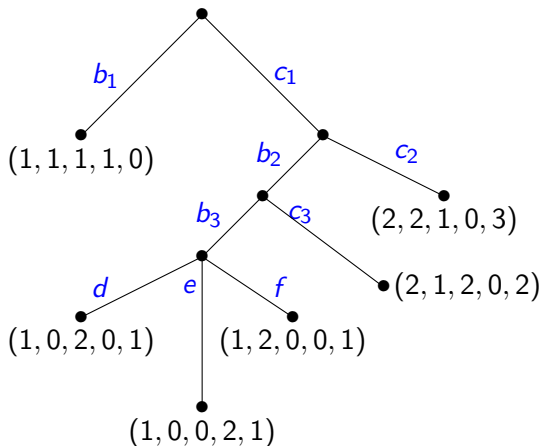
Caveat: The appropriate size restrictions must hold: any original set of mechanisms must be “large enough” to make sure that its image includes all payoff-relevant decisions .

A Sequential Environment (Calzolari-Pavan, 2008)

Four principals contract sequentially with the same agent: $\Theta = \{\theta\}$, and $E = \{e\}$. Take $D_j = \{b_j, c_j\}$ for $j = 1, 2, 3$ and $D_4 = \{d, e, f\}$:

- P1's mechanism γ_1 is NOT observed by downstream principals
- All other mechanisms and the agent's reports in any such mechanism are observed by downstream principals
- We identify a mechanism with its image, i.e. γ_i^{xy}

A Sequential Environment: Payoffs



A Sequential Environment: Feasible Mechanisms

- $\Gamma_1 = \{\gamma_1^{b_1}, \underline{\gamma}_1^{c_1}, \overline{\gamma}_1^{c_1}, \gamma_1^{b_1 c_1}\}$
- $\Gamma_i = \{\gamma_i^{b_i}, \gamma_i^{c_i}, \gamma_i^{b_i c_i}\}$ for $i = 2, 3$
- $\Gamma_4 = \{\gamma_4^d, \gamma_4^e, \gamma_4^f, \gamma_4^{de}, \gamma_4^{df}, \gamma_4^{ef}, \gamma_4^{def}\}$

A Sequential Environment: PBE

- ① P1 posts $\gamma_1^{b_1}$
- ② P2 posts $\gamma_2^{b_2}$ for each d_1
- ③ P3 posts $\gamma_3^{b_3}$ for each γ_2 and for each (d_1, d_2)
- ④ P4 posts γ_4^{def} if (c_1, b_2, b_3) and γ_4^d otherwise

Observing $d_1 = c_1$ but without observing γ_1 :

P2 believes $\bar{\gamma}_1^{c_1}$, P3 believes $\gamma_1^{b_1 c_1}$, P4 believes $\underline{\gamma}_1^{c_1}$

The Agent's Equilibrium Strategy (Incomplete)

- $t=1 \rightarrow d_1 = c_1$ if $\gamma_1^{b_1 c_1}$
- $t=2 \rightarrow d_2 = c_2$ if $\gamma_2^{b_2 c_2}$
- $t=3 \rightarrow d_3 = c_3$ if $\gamma_3^{b_3 c_3}$
- $t=4 \rightarrow d_4 = e$ if $\underline{\gamma}_1^{c_1}$, $d_4 = f$ if $\bar{\gamma}_1^{c_1}$, $d_4 = d$ if $\{\gamma_1^{b_1}, \gamma_1^{b_1 c_1}\}$

At $t = 4$, we are assuming that these decisions are made available by γ_4

Equilibrium Analysis

On-path decisions (b_1, b_2, b_3, d) . Payoffs: $(1, 1, 1, 1, 0)$

Following a deviation of P1:

- ① P2 does NOT induce c_2 because he believes that the agent will choose f
- ② P3 does NOT induce c_3 because he believes that the agent will choose d
- ③ P4 does NOT offer $\{e\}$ because he believes that the agent will choose e in $\{d, e, f\}$

Failure of the Delegation Theorem (1)

Consider the “Menu Game” $\hat{G}$:

- ① P1 does NOT deviate to $\{c_1\} \rightarrow d_2 = b_2$ and $d_3 = b_3$
- ② P2 does NOT deviate to $\{c_2\} \rightarrow d_3 = b_3$ and $d_4 = f$
- ③ P3 does NOT deviate to $\{c_3\} \rightarrow d_4 = d$
- ④ P3 does NOT deviate to $\{e\} \rightarrow d_4 = e$

Failure of the Delegation Theorem (2)

Given A's preferences:

- ① P2 posts $\{b_2\}$ if $d_1 = c_1$
- ② P3 posts $\{b_3\}$ if $d_1 = c_1 \quad a_2 = b_2$
- ③ P4 posts $\{def\}$ if $d_1 = c_1 \quad a_2 = b_2 \quad a_3 = b_3$

P2 P3 and P4 NEED to have different beliefs on the menu posted by P1

Yet there are only TWO menus containing $c_1 \rightarrow$ contradiction

Contractible Contracts (Kalai-Kalai-Lehrer-Samet 2010)

Two-player finite games $G = \{N = 2, (S_1, S_2), U : S \rightarrow \mathbb{R}^2\}$

A mixed strategy for player i is $\sigma_i \in \Delta S_i$: $\sigma(s) = \sigma_1(s_1)\sigma_2(s_2)$

A correlated strategy is $z \in \Delta S$

z is *IR* if $U_i(z) \geq \min_{\sigma_{-i}} \max_{\sigma_i} U_i(\sigma)$ for each i

Device Space

A device for player i is $d_i \in D_i$

d_i induces $r_{d_i} : D_{-i} \rightarrow S_i$, the strategy that d_i selects for i against D_{-i}

$$R(d_1, d_2) = (r_{d_1}(d_2), r_{d_2}(d_1))$$

Device Space: $(D, R) = \{(D_1, D_2), (R_1, R_2)\}$ with $R : D \rightarrow S$

Device Space

A device space (D, R) induces a two-player commitment game G^D :

- A (pure) strategy for player i is $d_i \in D_i$
- The payoff to player i is $U_i(R(d))$

Definition: A commitment-device equilibrium is a device space D and a Nash equilibrium of G^D .

Remark: Appropriate measurability assumptions should be made to deal with infinite device spaces.

Voluntary Device Spaces

Definition: The device space D is voluntary for player i if, for every $s_i \in S_i$, there is a $s_i^D \in D_i$ such that for every $d_{-i} \in D_{-i}$:

❶ (Unconditioned play): $r_{s_i^D}(d_{-i}) = s_i$

❷ (Private play): for every pair $(s_i^D, \tilde{s}_i^D)$, we have $r_{d_{-i}}(s_i^D) = r_{d_{-i}}(\tilde{s}_i^D)$

Voluntary device spaces allow to play the game G as scheduled, with no additional commitment.

Player i can choose any $s_i \in S_i$ without conditioning on the opponent choices and with the opponent unable to condition on s_i .

Example: Divorce Settlement

Two players are involved in a divorce settlement: $S_1 = S_2 = \{c, a\}$

	c	a
c	3, 3	0, 4
a	4, 0	1, 1

Each player can hire a lawyer who could be *flexible* (FL) or *tough* (TL):

- A *TL* lawyer always chooses $\{a\}$
- A *FL* lawyer chooses $\{c\}$ only if the opponent is *FL* and $\{a\}$ otherwise

Divorce Settlement: the G^D Game

$D_1 = D_2 = \{FL, TL, c^D, a^D\}$ and (R_1, R_2) are such that:

	FL	TL	c^D	a^D
FL	(c, c)	(a, a)	(a, c)	(a, a)
TL	(a, a)	(a, a)	(a, c)	(a, a)
c^D	(c, a)	(c, a)	(c, c)	(c, a)
a^D	(a, a)	(a, a)	(a, c)	(a, a)

(FL, FL) is an equilibrium of G^D

Remark: equilibrium strategies resemble tit-for-tat in a one-shot game

A Folk Theorem

Definition: A space of commitment devices is complete if the payoffs of every (IR) correlated strategy obtain at some (possibly mixed) equilibrium of G^D

Theorem (Kalai-Kalai-Lehrer-Samet): For any finite 2-player game there is a complete, voluntary space of devices

Example: Commitment vs Correlated Equilibrium

Two players choose to "Fight" or "Relinquish": $S_1 = S_2 = \{F, R\}$

	F	R
F	2, 2	10, 0
R	0, 10	0, 0

(F, F) is the unique correlated equilibrium.

min-max strategies guarantee each player a payoff of 2. The above Folk theorem suggests that in a commitment-device equilibrium:

Every payoff in the convex hull of $\{(2, 2), (2, 8), (8, 2)\}$ can be supported at equilibrium

Competing Mechanisms: The three pillars of wisdom

1. Failure of the (standard) Revelation Principle: Restricting principals to direct mechanisms that implement a decision for every profile of reported types is with loss of generality (McAfee 1993, Peck 1997)
2. Equilibrium mechanisms: Epstein-Peters (1999) identify **universal message spaces** allowing to reformulate a revelation principle
3. Equilibrium allocations: If agents' message sets are rich enough, a **folk theorem** obtains: every IC allocation that yields every principal a payoff above a min-max-min bound can be supported in equilibrium (Yamashita, 2010, Peters-Troncoso, 2013)

Games with standard mechanisms

$J \geq 2$ principals and $I \geq 2$ agents

- Principals post **public** mechanisms $\phi_j : M_j \equiv \prod_i M_j^i \rightarrow \Delta(D_j)$
- Agents learn their (exogenous) types ω^i
- Agents send **private** messages to each principal
- Principals' decisions are implemented

The standard mechanisms game G^M

What is not captured? Private disclosures!

- Each principal may privately disclose information to agents **before** their messages are sent
- Different agents may be differently informed about the principal's allocation rule
- This is an additional source of incomplete information in the agents' message game
- The “market information” may therefore be asymmetric between agents

Mechanisms with private disclosures

- Let S_j^i be the set of private disclosures of principal j to agent i
- Denote $S_j \equiv \prod_i S_j^i$
- A mechanism with private disclosures for principal j consists of a joint distribution over S_j and an allocation rule

$$\sigma_j \in \Delta(S_j), \quad \phi_j : S_j \times M_j \rightarrow \Delta(D_j)$$

- Each signals profile $s_j \equiv (s_j^1, \dots, s_j^I)$ induces a standard mechanism $\phi_j(s_j) : M_j \rightarrow \Delta(D_j)$
- The signal s_j^i identifies the private information of agent i about principal j 's allocation rule

Competing mechanisms with private disclosures

- 1 Principals post public mechanisms

$$\sigma_j \in \Delta(S_j), \quad \phi_j : S_j \times M_j \rightarrow \Delta(D_j)$$

- 2 Agents learn their (exogenous) types ω^i and receive **private** disclosures $(s_1^i, \dots, s_j^i) \in S_1^i \times \dots \times S_j^i$
- 3 Agents send **private** messages to each principal
- 4 Principals' decisions are implemented

This is the game G^{SM} with private disclosures

Equilibrium

Let Γ_j be the space of admissible mechanisms for principal j and $\Gamma \equiv \prod_j \Gamma_j$

The strategy profile (μ, λ) is a PBE of G^{SM} whenever

- ① for each $\gamma \in \Gamma$, $(\lambda^1(\gamma), \dots, \lambda^I(\gamma))$ is a BNE of the subgame γ played by the agents
- ② given the continuation equilibrium strategies λ , μ is a NE of the game played by the principals

One vs Several Principals

One Principal

- Private disclosures have no impact on the set of implementable allocations. No loss if the principal publicly communicates via the mechanism she commits to and agents report their types
- Agents may hold private information over the allocation rule **if the principal cannot commit to a “grand” public mechanisms**: Hart-Tirole (1990), McAfee-Schwartz (1994), Segal-Whinston (2003)

Several Principals a principal may **choose** to privately inform agents in order to discipline their communication with his opponents

Results: undermining the pillars

- Exhibit equilibrium allocations of G^M games that cannot be supported when disclosures are allowed, **no matter how rich M is**
- Exhibit equilibrium allocations of G^{SM} games that cannot be supported in G^M , **no matter how rich M is**
- Identify a **canonical** game G^{SM} :
 - Enriching communication does not add equilibria
 - The equilibria of G^{SM} are robust

A Folk Theorem in Recommendation Mechanisms

Theorem (Yamashita 2010) *If $I \geq 3$, every IC allocation yielding each principal a payoff above a well specified bound is supportable in equilibrium*

The result exploits the possibility for a principal to have agents “recommending” him the state-contingent decision to implement against his opponents’ deviations

We exhibit equilibrium allocations supported by standard mechanisms—regardless of the size of M —that **fail to be equilibria in games with private disclosures**

Example 1: Game form

- Two principals, P1 and P2, three agents A1, A2, and A3
- P1's decision in $D_1 \equiv \{x_1, x_2\}$, P2's decision in $D_2 \equiv \{y_1, y_2\}$
- A1's and A2's exogenous type in $\Omega^1 = \Omega^2 \equiv \{\omega_L, \omega_H\}$
- A3 has no exogenous private information
- A1's and A2's types are perfectly correlated

Example 1: Payoffs

- P1's and A3's payoffs are constant across decisions and states
- Payoffs for P2, A1, and A2 are given by

$$\omega = (\omega_L, \omega_L)$$

	y_1	y_2
x_1	5, 8, 8	5, 1, 1
x_2	6, 4.5, 4.5	6, 4.5, 4.5

$$\omega = (\omega_H, \omega_H)$$

	y_1	y_2
x_1	6, 4.5, 4.5	6, 4.5, 4.5
x_2	5, 1, 1	5, 8, 8

The game with standard mechanisms G^M

- $t = 0$ Principals simultaneously post mechanisms ϕ_j
- $t = 1$ A1 and A2 learn ω^1 and ω^2 , respectively
- $t = 2$ Agents send messages m_j^i
- $t = 3$ Principals' decisions are implemented according to $\phi_j(m_j)$

A folk-theorem result (Yamashita, 2010)

Lemma

Suppose $M_j^i \supseteq D_j \times \Omega^i$ for all i and j , with each M_j^i finite. Then any payoff for $P2$ in the feasible set $[5, 6]$ can be supported in a PBE of G^M .

Here we focus on the minimal equilibrium payoff of 5

$\omega = (\omega_L, \omega_L)$			$\omega = (\omega_H, \omega_H)$		
	y_1	y_2		y_1	y_2
x_1	5, 8, 8	5, 1, 1	x_1	6, 4.5, 4.5	6, 4.5, 4.5
x_2	6, 4.5, 4.5	6, 4.5, 4.5	x_2	5, 1, 1	5, 8, 8

Lemma 1': On path

$\omega = (\omega_L, \omega_L)$			$\omega = (\omega_H, \omega_H)$		
	y_1	y_2		y_1	y_2
x_1	5, 8, 8	5, 1, 1	x_1	6, 4.5, 4.5	6, 4.5, 4.5
x_2	6, 4.5, 4.5	6, 4.5, 4.5	x_2	5, 1, 1	5, 8, 8

- P1 and P2 post recommendation mechanisms (ϕ_1^r, ϕ_2^r)
- In the subgame (ϕ_1^r, ϕ_2^r) , all agents recommend

$$d_1^*(\hat{\omega}) \equiv \begin{cases} x_1 & \text{if } \hat{\omega} = (\omega_L, \omega_L) \\ x_2 & \text{otherwise} \end{cases} \quad d_2^*(\hat{\omega}) \equiv \begin{cases} y_1 & \text{if } \hat{\omega} = (\omega_L, \omega_L) \\ y_2 & \text{otherwise} \end{cases}$$

and A1 and A2 report truthfully to both principals

Lemma 1': Off path

- Suppose P2 deviates to $\phi_2 : M_2 \rightarrow \Delta(D_2)$
- Let $p(m_2) \equiv \Pr[y_1 | m_2]$

$$\bar{p} \equiv p(\bar{m}_2) \geq p(m_2) \quad \forall m_2$$

$$\underline{p} = p(\underline{m}_2^1, \underline{m}_2^2, \bar{m}_2^3) \leq p(m_2^1, m_2^2, \bar{m}_2^3) \quad \forall (m_2^1, m_2^2)$$

Lemma 1': Off path

$\omega = (\omega_L, \omega_L)$			$\omega = (\omega_H, \omega_H)$		
	y_1	y_2		y_1	y_2
x_1	5, 8, 8	5, 1, 1	x_1	6, 4.5, 4.5	6, 4.5, 4.5
x_2	6, 4.5, 4.5	6, 4.5, 4.5	x_2	5, 1, 1	5, 8, 8

Case 1 $\bar{p} \geq \frac{1}{2}$

- All agents recommend to P1 $d_1^*(\hat{\omega}) \equiv \begin{cases} x_1 & \text{if } \hat{\omega} = (\omega_L, \omega_L) \\ x_2 & \text{otherwise} \end{cases}$
- A3 sends $\bar{m}_2^3$ to P2
 - (ω_L, ω_L) : $8\bar{p} + (1 - \bar{p}) \geq 4.5 \implies$ truthful reporting + $\bar{m}_2^i$ optimal
 - (ω_H, ω_H) : no agent can unilaterally change P1's decision
- Truthful reports to P1 $\implies$ P2's payoff is 5

Lemma 1': Off path

$\omega = (\omega_L, \omega_L)$			$\omega = (\omega_H, \omega_H)$		
	y_1	y_2		y_1	y_2
x_1	5, 8, 8	5, 1, 1	x_1	6, 4.5, 4.5	6, 4.5, 4.5
x_2	6, 4.5, 4.5	6, 4.5, 4.5	x_2	5, 1, 1	5, 8, 8

Case 2 $\bar{p} < \frac{1}{2}$

- All agents recommend to P1 $d_1(\hat{\omega}) \equiv \begin{cases} x_2 & \text{if } \hat{\omega} = (\omega_H, \omega_H) \\ x_1 & \text{otherwise} \end{cases}$
- A3 sends $\bar{m}_2^3$ to P2
 - (ω_L, ω_L) : no agent can unilaterally change P1's decision
 - (ω_H, ω_H) : $\underline{p} + 8(1 - \underline{p}) \geq 4.5 \implies$ truthful reporting + $\underline{m}_2^i$ optimal
- Truthful reports to P1 $\implies$ P2's payoff is 5

The game with private disclosures G^{SM}

- $t = 0$ Principals simultaneously post mechanisms (σ_j, ϕ_j)
- $t = 1$ A1 and A2 learn ω^1 and ω^2 , respectively, and receive signals s_j^i according to σ_j
- $t = 2$ Agents send messages m_j^i
- $t = 3$ Principals' decisions are implemented according to $\phi_j(s_j, m_j)$

The Mechanism $\bar{\gamma}_2$ yielding P2 more than 5

Assume $S_2^1 \equiv \{1, 2\}$ and let $\bar{\gamma}_2$ be the mechanism that

- does not respond to the agents' messages
- sends no signals to A2 and A3
- with probability $\sigma \in (\frac{1}{2}, 1)$ sends $s_2^1 = 1$ to A1 and selects y_1
- with probability $1 - \sigma$ sends $s_2^1 = 2$ to A1 and selects y_2

Lemma 2: P1's mechanism γ_1

$\omega = (\omega_L, \omega_L)$			$\omega = (\omega_H, \omega_H)$		
	y_1	y_2		y_1	y_2
x_1	5, 8, 8	5, 1, 1	x_1	6, 4.5, 4.5	6, 4.5, 4.5
x_2	6, 4.5, 4.5	6, 4.5, 4.5	x_2	5, 1, 1	5, 8, 8

- P2's payoff = 5 $\implies$ P1 must select x_1 in (ω_L, ω_L) and x_2 in (ω_H, ω_H)
- Decisions implemented in $\bar{\gamma}_2$ invariant to $m_2 \implies$ wlog γ_1 involves no signals
- (ω_L, ω_L) : after receiving $s_2^1 = 2$, A1 wants to minimize $\Pr[x_1]$
- (ω_H, ω_H) : after receiving $s_2^1 = 1$, A1 wants to maximize $\Pr[x_1]$

$\rightarrow \gamma_1$ must be such that A1 does not affect P1's decision in either state

$\rightarrow$ Because A3 does not know the state and P2 gets 5 only when P1's decision is perfectly correlated with the state, A2 must have full control over P1's decision

$\rightarrow$ Because $\sigma > \frac{1}{2}$, in state (ω_H, ω_H) A2 has a profitable deviation, which is to implement x_1 instead of x_2

The benefits of private disclosures

- The information P2 discloses to A1 makes A1 behave in P1's mechanism favorably to P2
- If P2 were to disclose the same information to A2 and A3, the agents could discipline each other and implement IC punishments for P2
- Private disclosures impede the agents' ability to coordinate on an appropriate punishment for P2 using P1's mechanism

The Non-Robustness (Anti-Folk-Theorem) Result

Proposition

Private disclosures allow P2 to defend a payoff above the minimal one of 5 in each subgame

Equilibrium outcomes of games in which principals are restricted to standard mechanisms need not be supportable when principals can engage in private disclosures

Example 2: Non-Universality

We identify an equilibrium allocation that can be supported using mechanisms with private disclosures, but **cannot** be supported in any G^M game

This challenges the canonicity of Epstein and Peters' (1999) universal mechanisms

Example 2: Insights

In the example, private disclosures are used **on path** to correlate the principals' decisions with the agents' types in a way that cannot be achieved with standard mechanisms

Private disclosures need not modify the agents' beliefs

They can be **encryption keys**: in isolation, each key is uninformative of a principal's decision but, taken together, they perfectly reveal it

Example 2: the Limits of Standard Mechanisms

In any G^M game:

- The equilibrium correlation can only be induced by the players' independent mixing: principals randomizing over mechanisms, and/or agents randomizing over the messages they send in those mechanisms
- In either case, the correlation over principals' decisions must ultimately obtain as an implication of incentive-compatible choices by the agents
- We show that, in this restricted game, and regardless of the source of randomness, there exist no agents' IC strategies inducing that correlation

Example 2: Game form

- Two principals, P1 and P2, two agents, A1 and A2
- P1's decisions in $A_1 \equiv \{x_1, x_2, x_3, x_4\}$, P2's decisions in $A_2 \equiv \{y_1, y_2\}$
- A1 has no exogenous private information
- A2's exogenous type in $\Omega^2 \equiv \{\omega_L, \omega_H\}$, with $\Pr[\omega^2 = \omega_L] = \frac{1}{4}$

Example 2: Payoffs

- P1's payoff is constant across decisions and states
- Payoffs for P2, A1, and A2 are given, for some $\zeta < 0$, by

$$\omega^2 = \omega_L$$

	y_1	y_2
x_1	$\zeta, 4, 1$	$\zeta, 8, 3.5$
x_2	$\zeta, 2, 5$	$\zeta, 9, 8$
x_3	$10, 3, 3$	$\zeta, 5.5, 3.5$
x_4	$\zeta, 1, 3.5$	$10, 7.5, 7.5$

$$\omega^2 = \omega_H$$

	y_1	y_2
x_1	$\zeta, 1, 6$	$10, 7.5, 5$
x_2	$10, 3, 9$	$\zeta, 5.5, 6$
x_3	$\zeta, 8, 7$	$\zeta, 4.5, 7$
x_4	$\zeta, 9, 6$	$\zeta, 3, 9$

Example 2: Signal and message sets

P2 sends signals to both agents but does not solicit messages

- Signal sets $S_2^1 = S_2^2 \equiv \{1, 2\}$, no messages

P1 sends no signals but asks for P2's signals and ω^2

- No signals, message sets $M_1^1 \equiv S_2^1$, $M_1^2 \equiv S_2^2 \times \Omega^2$

New equilibria supported by private disclosures

Proposition

*Games with private disclosures admit equilibrium allocations and payoffs that cannot be supported in **any** game without private disclosures.*

This holds regardless of

- how rich M is
- whether **public** randomizing devices are available

A “Failure” of the Delegation Principle (1)

Let $K = 2$, $N = 1$, $\Theta = \{\theta\}$, and $A = \{a\}$

Take also $D^1 = \{A, B\}$, $D^2 = \{A, B, C, D\}$. Payoffs are:

	A	B	C	D
A	$(-1, 0, 3)$	$(3, 0, 3)$	$(3, 0, 1)$	$(3, 0, 1)$
B	$(0, 0, 3)$	$(0, 0, 3)$	$(2, 2, 1)$	$(0, 0, 1)$
C	$(0, 0, 3)$	$(2, 1, 3)$	$(0, 0, 1)$	$(0, 0, 1)$

A “Failure” of the Delegation Principle (2)

Consider the following G^{SM} game:

$$S_1 = \{s\}, S_2 = \{\underline{s}_2, \bar{s}_2\}, M_1 = \{a, b, c\}, M_2 = \{a, b, c, d\}$$

Claim 1: The random allocation (C, B) with prob. $1/2$, (B, C) with prob. $1/2$ can be supported at equilibrium

A “Failure” of the Delegation Principle (3)

Consider the following mechanisms

$$\text{P1 posts } \gamma_1 = \phi_1 = \begin{cases} B & \text{if } m \in \{a, b\} \\ C & \text{if } m = c \end{cases}$$

P2 posts $\gamma_2 = (\sigma_2, \phi_2)$ with

$$\sigma_2(\underline{s}) = \sigma_2(\bar{s}) = 1/2$$

$$\phi_2(m, \underline{s}) = \begin{cases} A & \text{if } m \in \{a, b\} \\ B & \text{if } m \in \{c, d\} \end{cases} \quad \phi_2(m, \bar{s}) = \begin{cases} C & \text{if } m \in \{a, b\} \\ D & \text{if } m \in \{c, d\} \end{cases}$$

A “Failure” of the Delegation Principle (4)

On path: A chooses $\begin{cases} (C, B) & \text{if } \underline{s} \\ (B, C) & \text{if } \bar{s} \end{cases}$

Off path: We consider deviations from P1 and P2

P1 gets a payoff of 2. Suppose he offers the take-it-or-leave-it $\{A\}$. The agent's equilibrium strategy can be constructed to guarantee that she chooses $\{A\}$ in $\phi_2(m, \underline{s})$, which blocks the deviation.

P2 gets a payoff of 1.5. To profitably deviate, he would need to implement $\{C\}$ with probability $> 1/2$. The agent's equilibrium strategy can be constructed to prevent this, thereby blocking the deviation



A “Failure” of the Delegation Principle (5)

Claim 2: The allocation cannot be supported in an equilibrium of G^M , regardless of the size of M

Proof. This follows standard arguments based on principals' independent randomizations in G^M ■